

Figure 1: Directory structure and files in Instant Contiki

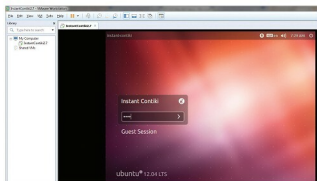


Figure 2: Instant Contiki login screen

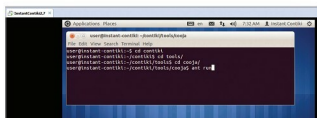


Figure 3: Loading the Cooja simulator in Contiki

Implementations for engineering

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Using the Contiki OS with the Cooja simulator to program the IoT

Contiki (<http://www.contiki-os.org/>) is a popular, free and open source operating system for IoT programming available under the BSD licence with the base code of the C programming language. Contiki can be used for communication between low powered RFID chips in wireless networks with a high degree of performance and security.

The programming on Contiki is done using the Cooja network simulator, in which the base libraries of RFID chips and sensors are available in C. To program, control and monitor the remote IoT devices, the back-end C programs and related header files can be customised and recompiled

to get the desired results. Contiki works on IPv4 as well as IPv6 networking with the integration of lightweight protocols, so that low power chips and radio frequency chips can be connected without performance issues.

The URL for downloading Instant Contiki is <http://sourceforge.net/projects/contiki/files/Instant%20Contiki/>.

Once the compressed Instant Contiki is downloaded, it can be used on any host operating system. Instant Contiki is available on sourceforge.net as a compressed file which then has to be extracted. The uncompressed or extracted Instant Contiki can be executed on VMware Player, which is a virtualisation tool that can be downloaded free and is available at <http://www.vmware.com/go/downloadplayer/>.

In the extracted folder of Instant Contiki, there is the executable file `Instant-Contiki-Ubuntu_12.04_32-bit.vmx`.

This file, on executing, will automatically open in VMware Player, and we will be ready to work with Contiki using virtualisation software in parallel with any host operating system. The default password for the Contiki operating system is 'user'.

After loading the Contiki OS, the commands given in Figure 3 are executed in the terminal so that the Cooja simulator gets loaded for the implementation of IoT.

Creating a new network in the Cooja simulator

In the file menu of Cooja, select `New Simulation` as shown in Figure 4.

In the dialogue box, the basic network parameters are

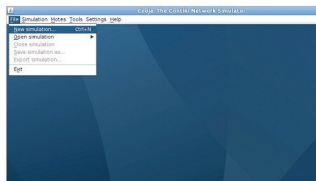


Figure 4: Creating new IoT simulation in Cooja

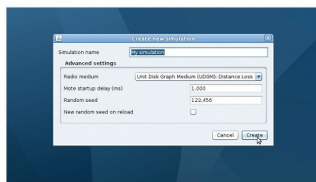


Figure 5: Set-up of basic simulation properties in Cooja